# Enhanced vascular-program detection · findings

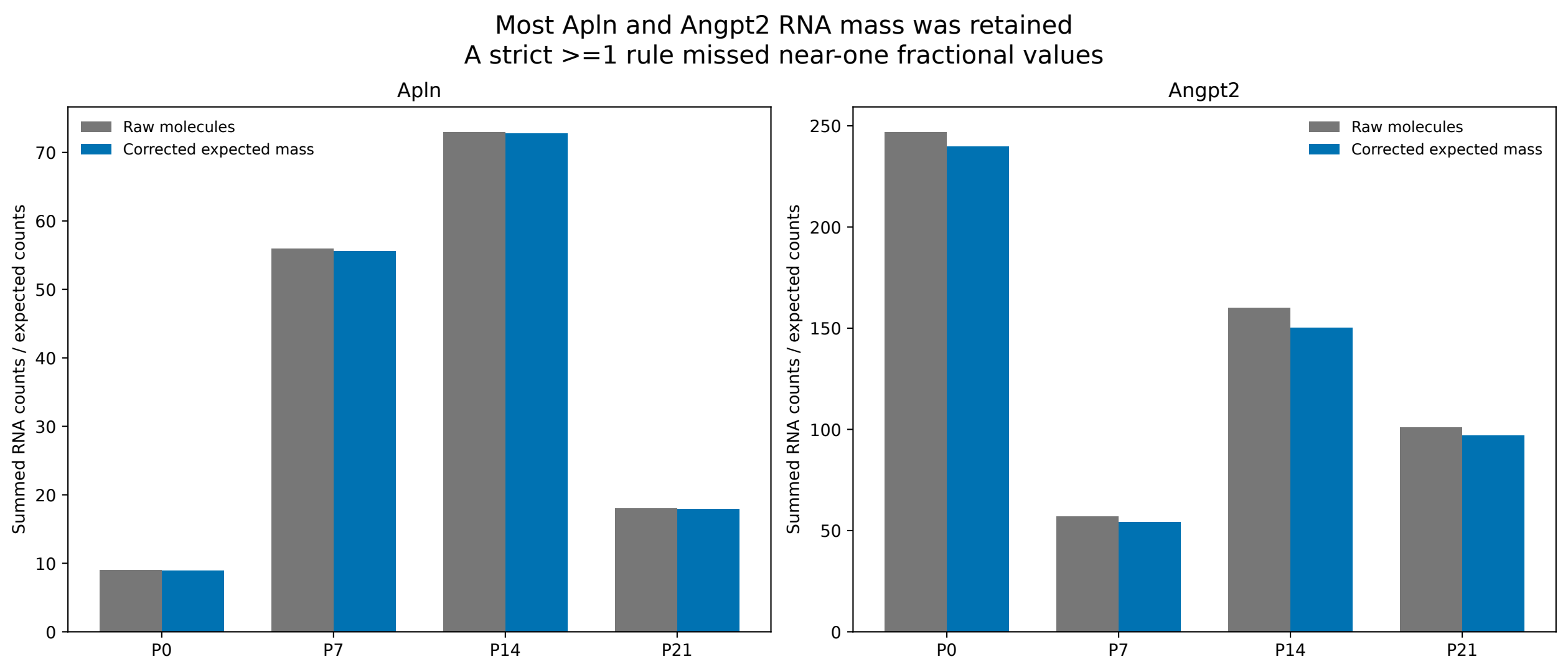
An important correction: applying an integer threshold to fractional DecontX counts understated retained low-count RNA. P7 Apln has56 raw molecules and55.6 corrected expected molecules. The old >=1 rule counted5 nuclei; raw-supported fractional thresholds recover47. This is a quantification correction, not restoration of removed ambient RNA.

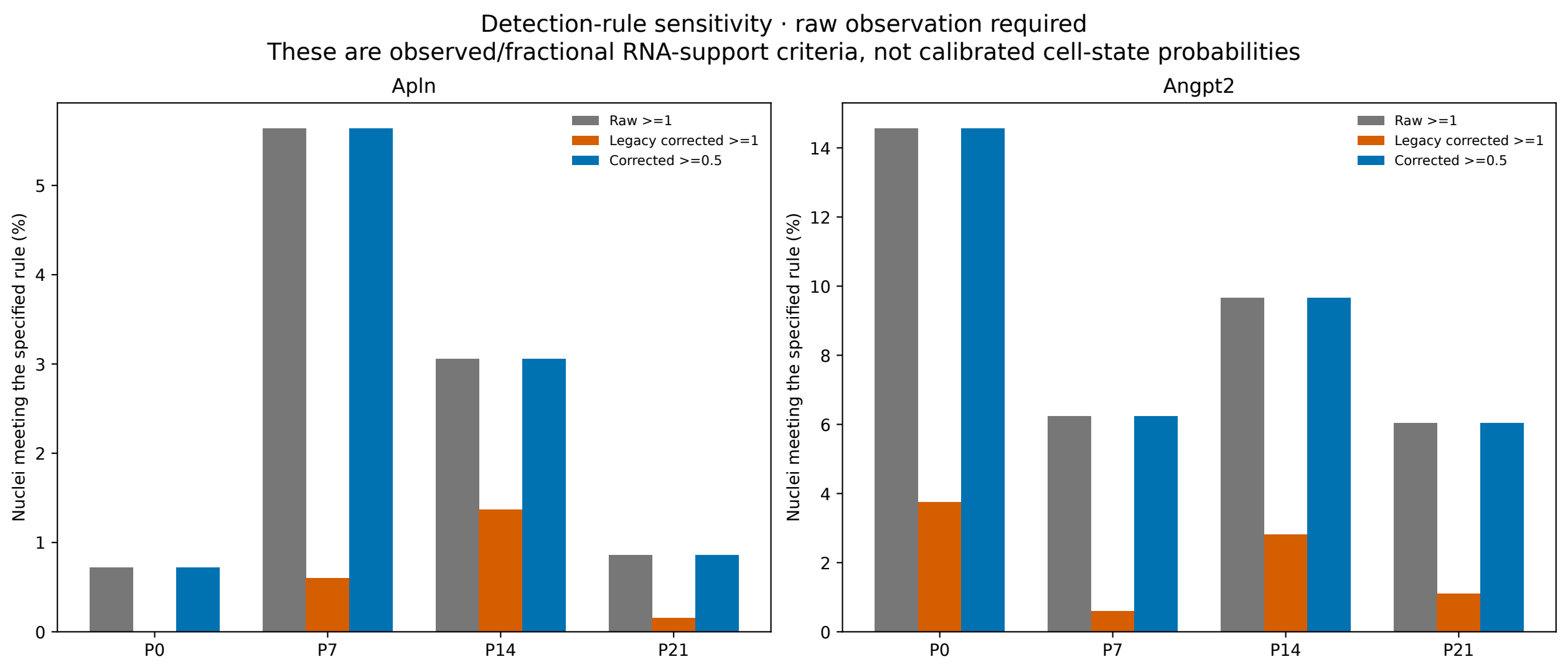
RNA enhancement: preserve fractional counts; report expected-mass pseudobulks and threshold sensitivity; use expression/detection-matched rank enrichment; require age-matched library support; audit EC versus non-EC raw RNA and gene coherence. A cellular-background estimate is only a sensitivity proxy because no empty-droplet background is available.

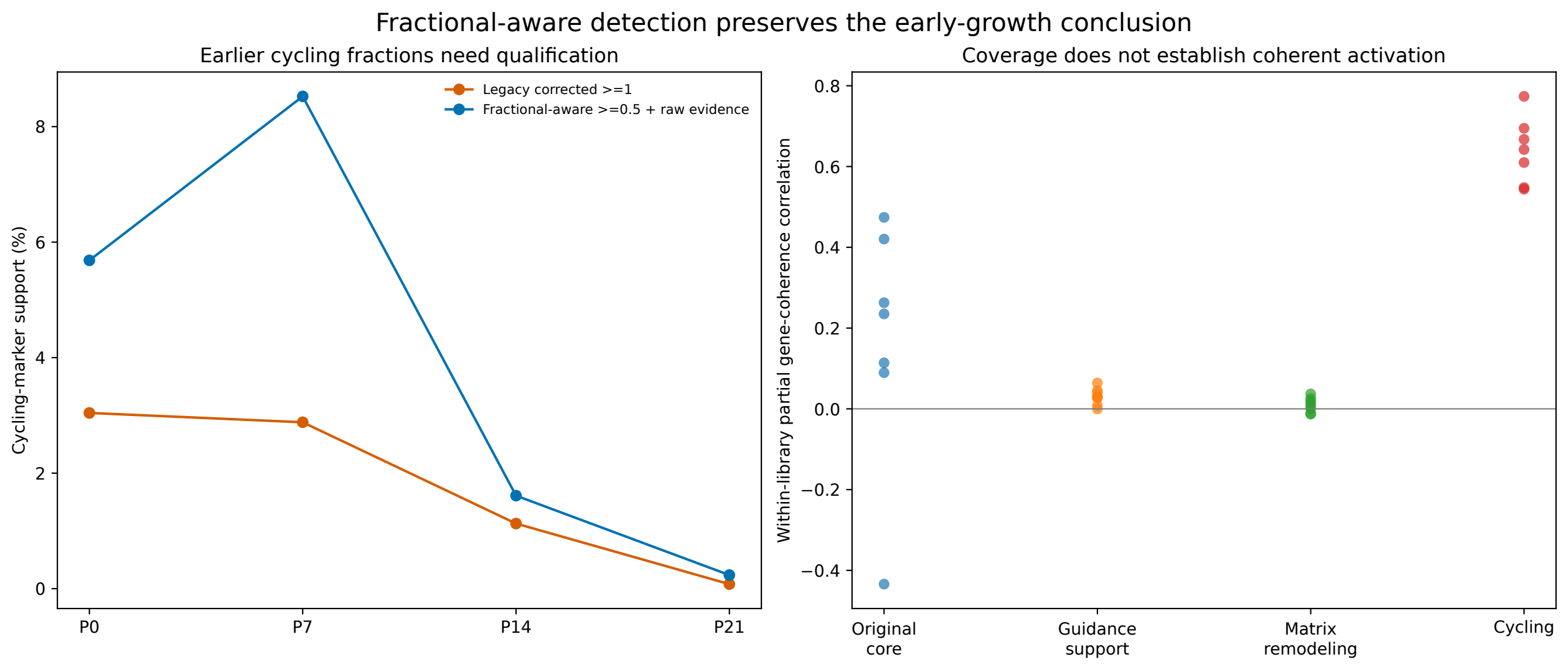
The earlier cycling percentages also used the strict integer rule. A fractional-aware >=0.5 expected-count threshold plus raw observation gives71/1249 atP0,71/833 atP7,20/1242 atP14 and3/1273 atP21. Thresholds0.1/0.25/0.5 agree here. These are marker-support fractions in the frozen QC cohort, not independently calibrated biological proliferation rates.

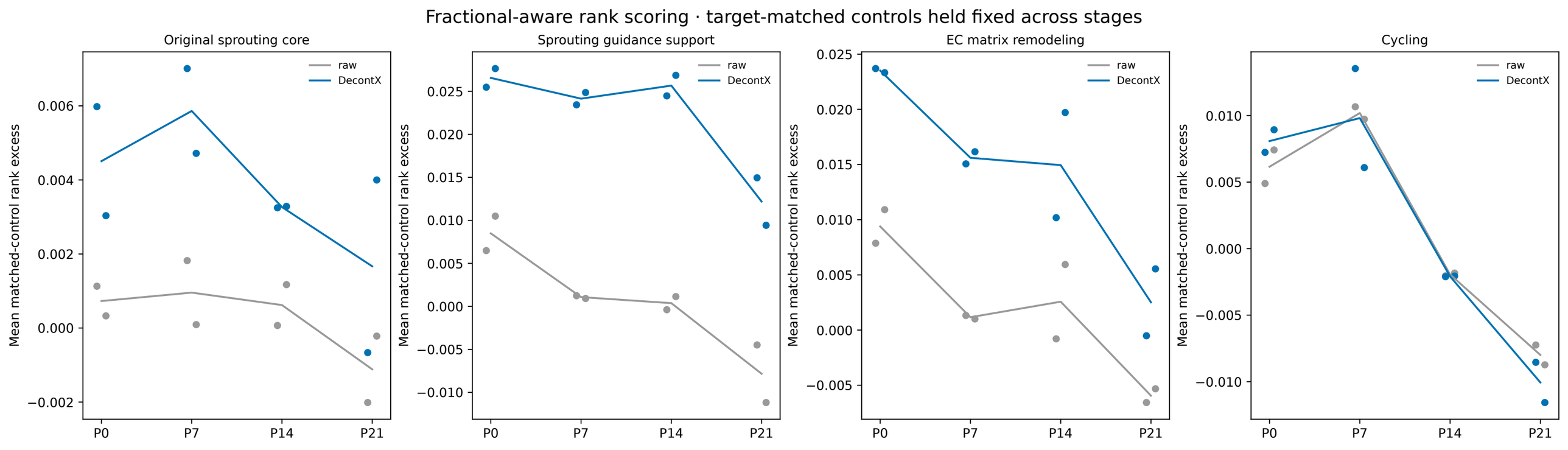
Xenium directly measures onlyAngpt2 from the original six-gene sprouting panel. Apln/Esm1/Cxcr4/Pgf/Adm are unmeasured. Observed guidance, matrix and cycling proxies are kept separate. Neither lower RNA-quality cutoffs nor invented counts add these missing genes.

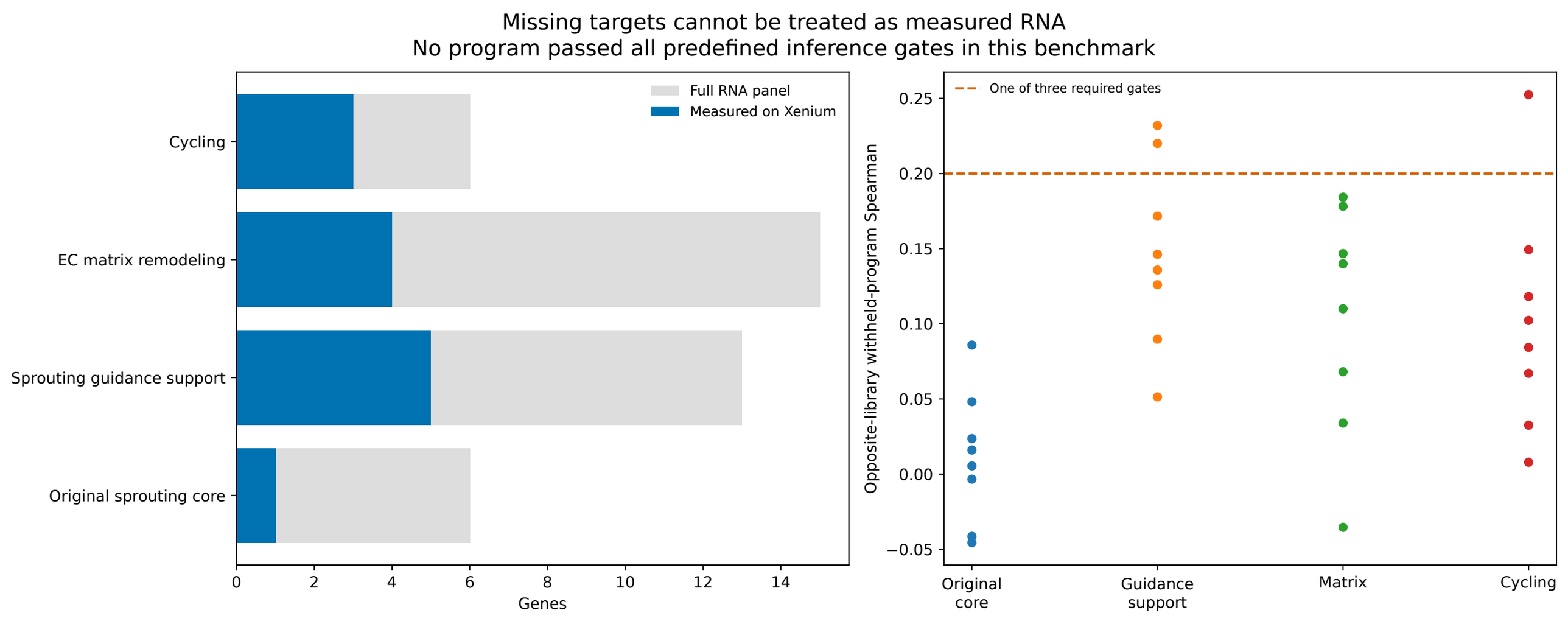
Spatial enhancement: frozen nuclear-clean ECs, nuclear/core/native and matched-depth controls, non-overlapping25/50/100um tiles, and five-EC pools within75um. The latter cover89-99% of the selected EC positions but overlap, so they are displays at neighborhood resolution, not independent observations or cell-specific states.

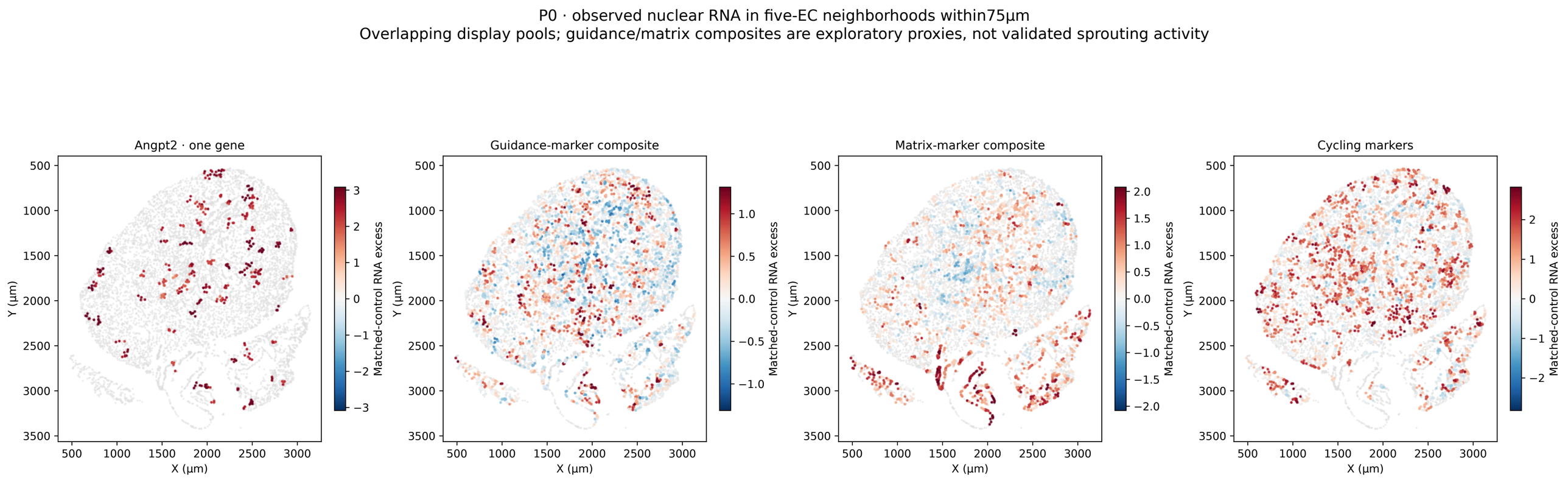

Most Apln and Angpt2 RNA mass was retained
A strict >=1 rule missed near-one fractional values
Apln
Angpt2
Raw molecules
Corrected expected mass
Summed RNA counts / expected counts
0
10
20
30
40
50
60
70
50
100
150
200
250
P0
P7
P14
P21

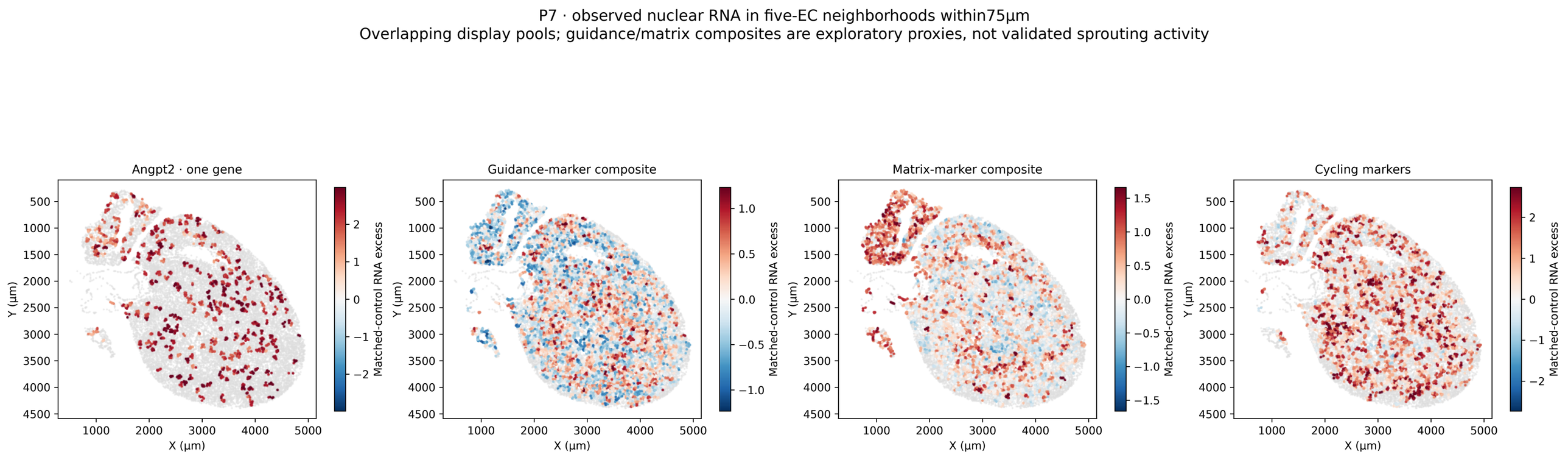

Detection-rule sensitivity · raw observation required
These are observed/fractional RNA-support criteria, not calibrated cell-state probabilities
Apln
Angpt2
Raw >=1
Legacy corrected >=1
Corrected >=0.5
Nuclei meeting the specified rule (%)
P0
P7
P14
P21

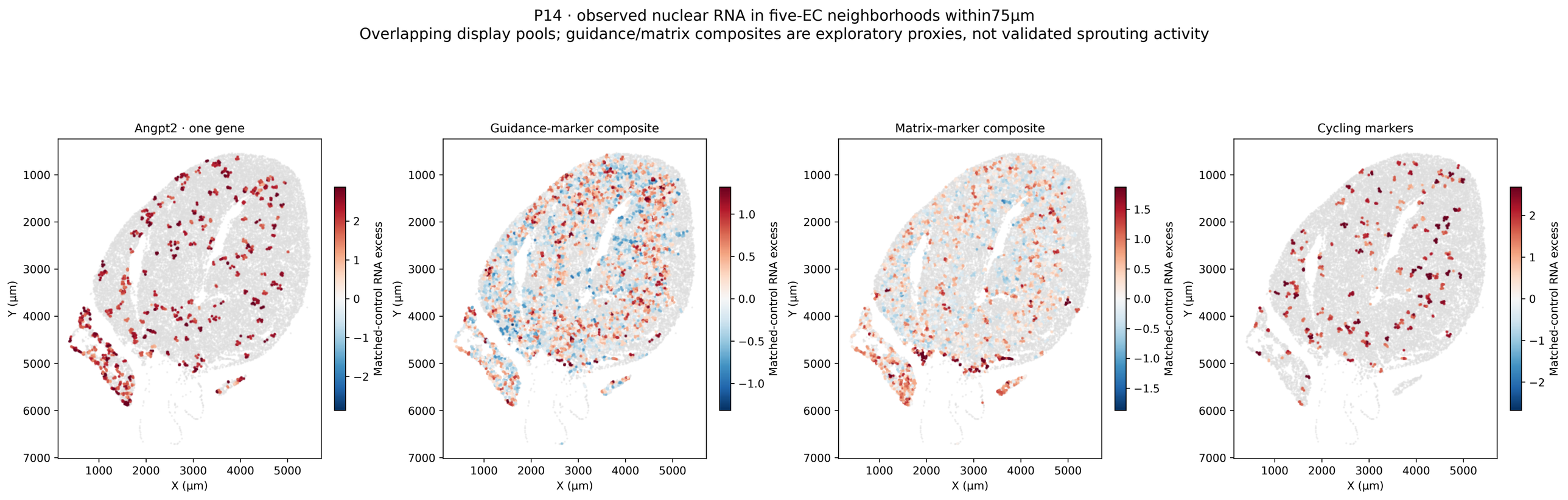

Fractional-aware detection preserves the early-growth conclusion
Earlier cycling fractions need qualification
Legacy corrected >=1
Fractional-aware >=0.5 + raw evidence
Cycling-marker support (%)
0
2
4
6
8
P0
P7
P14
P21
Coverage does not establish coherent activation
Within-library partial gene-coherence correlation
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
Original core
Guidance support
Matrix remodeling
Cycling

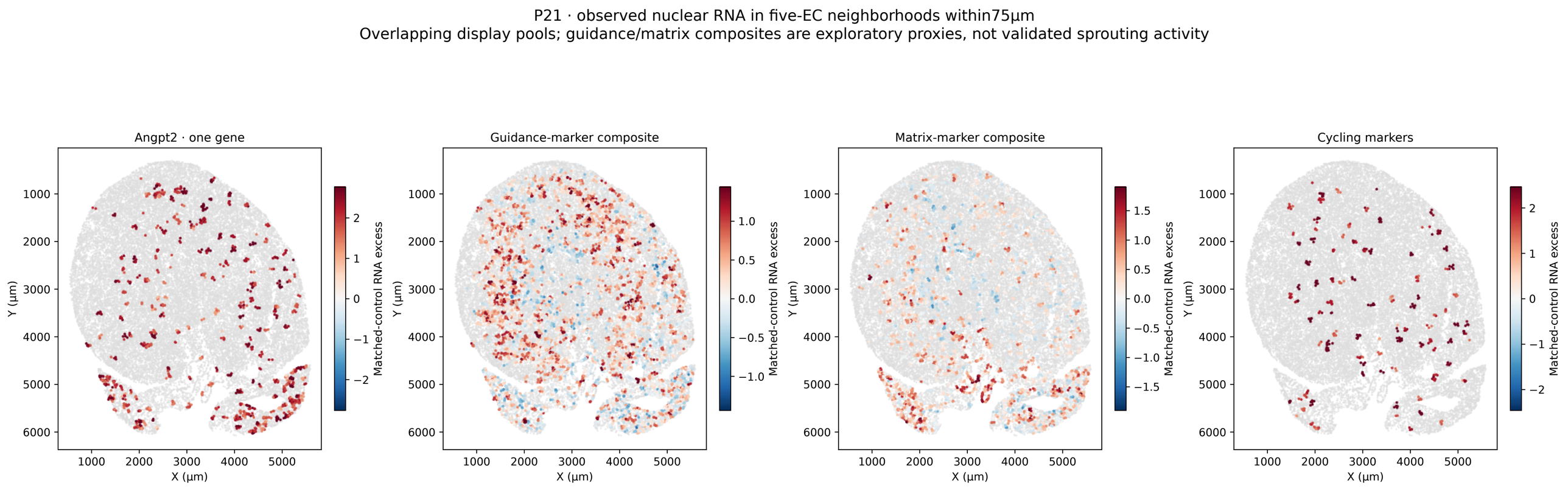

Fractional-aware rank scoring · target-matched controls held fixed across stages
Original sprouting core
Sprouting guidance support
EC matrix remodeling
Cycling
raw
DecontX
Mean matched-control rank excess
P0
P7
P14
P21


Missing targets cannot be treated as measured RNA
No program passed all predefined inference gates in this benchmark
Full RNA panel
Measured on Xenium
Cycling
EC matrix remodeling
Sprouting guidance support
Original sprouting core
0
2
4
6
8
10
12
14
Genes
One of three required gates
Opposite-library withheld-program Spearman
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
Original core
Guidance support
Matrix
Cycling


P0 · observed nuclear RNA in five-EC neighborhoods within75µm
Overlapping display pools; guidance/matrix composites are exploratory proxies, not validated sprouting activity
Angpt2 · one gene
Guidance-marker composite
Matrix-marker composite
Cycling markers
X (µm)
Y (µm)
Matched-control RNA excess


P7 · observed nuclear RNA in five-EC neighborhoods within75µm
Overlapping display pools; guidance/matrix composites are exploratory proxies, not validated sprouting activity
Angpt2 · one gene
Guidance-marker composite
Matrix-marker composite
Cycling markers
Y (µm)
X (µm)
Matched-control RNA excess


P14 · observed nuclear RNA in five-EC neighborhoods within75µm
Overlapping display pools; guidance/matrix composites are exploratory proxies, not validated sprouting activity
Angpt2 · one gene
Guidance-marker composite
Matrix-marker composite
Cycling markers
X (µm)
Y (µm)
Matched-control RNA excess


P21 · observed nuclear RNA in five-EC neighborhoods within75µm
Overlapping display pools; guidance/matrix composites are exploratory proxies, not validated sprouting activity
Angpt2 · one gene
Guidance-marker composite
Matrix-marker composite
Cycling markers
Y (µm)
X (µm)
Matched-control RNA excess

# Interpretation and remaining limits

Detection has improved without manufacturing RNA. Corrected expected mass and fractional-aware scoring recover low-count EC support that the earlier integer cutoff obscured. Raw and corrected matrices are unchanged; lowering a detection threshold does not identify true biology by itself.

Apln and Angpt2 have support in both libraries at several stages. Esm1 remains genuinely sparse in this RNA dataset; it is absent in raw P7 EC counts. The full six-gene program does not show uniformly strong within-library coherence. Expanded guidance and remodeling panels are supportive functional composites, not replacements for the original biological definition.

Spatial pooling increases the chance of observing multiple genes, at the cost of resolution. The five-EC displays overlap and can combine neighboring vessels; they must not be counted as independent active cells. Non-overlapping tiles, common source profiles and molecule-depth controls supply the more conservative sensitivity/coherence comparisons.

OnlyAngpt2 from the original six-gene panel is directly assayed spatially. This specific cross-library Ridge benchmark does not justify imputing missing program genes; it does not prove every possible inference method must fail. Direct Apln/Esm1 validation would require additional measured targets, such as targeted RNA in situ with EC and membrane/nuclear markers.

Non-EC raw cellular RNA is not measured ambient soup. Candidate overattenuation flags therefore cannot authorize restoring raw counts. The five vascular EC identity labels and the original reference cohort remain fixed; formerly reviewed ECs are evaluated only in a separate fractional-aware sensitivity table.

Two RNA libraries per age are not assumed independent animals. One Xenium section per age does not support organ-wide developmental rates or causal vessel growth. Improved coverage is not independently validated program accuracy.